# Sungjin Cheong

CS Ph.D. Student @ University of Wisconsin-Madison

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## Education

- **University of Wisconsin-Madison** — *Sep. 2025 – Present*
  Ph.D. in Computer Sciences (*Advisor: Mohit Gupta*) — Madison, WI, USA
- **KAIST** — *Sep. 2021 – Aug. 2023*
  M.S. in Mechanical Engineering (*Advisor: Yong-Hwa Park*) — Daejeon, Republic of Korea
- **Yonsei University** — *Mar. 2014 – Aug. 2021*
  B.S. in Mechanical Engineering — Seoul, Republic of Korea

## Research Interests

- Physics-grounded intelligent sensing and imaging systems
- ToF/LiDAR computational imaging
- Statistical signal processing and inverse problems
- Physics-based and learning-based modeling

## Publications & Conferences

J=Journal, IC=International Conference, DC = Domestic Conference

[IC.1] **Ghosts in the Point Clouds: De-glaring LiDAR in the Transient Domain**
Gump, A.; Henley, C. A.; **Cheong, S. J.**; Prabhakara, A.; Gupta, M. *CVPR*

[J.1] **Thermal-Infrared Remote-Target Detection System for Maritime Rescue Using 3-D Game-Based Data Augmentation With GAN**
**Cheong, S. J.**; Jung, W.; Lim, Y. S.; Park, Y. H. *IEEE Transactions on Geoscience and Remote Sensing (*IF: 9.4)*

[IC.2] **LiDAR Blooming Artifacts Estimation Method Induced by Retro-Reflectance With Synthetic Data Modeling and Deep Learning**
**Cheong, S. J.**; Ha, J. S. *ICCE-Asia*

[IC.3] **Infrared Image-Based Remote Target Detection for Maritime Rescue Utilizing a Deep Learning Network and Data Augmentation**
**Cheong, S. J.**; Lim, Y. S.; Jung, W. H.; Park, Y. H. *SPIE*

[IC.4] **MEMS-Based Indirect Time-of-Flight Scanning LiDAR With Parallel-Phase Demodulation and Multi-path Interference Suppression**
Lee, S. H.; Lim, Y. S.; **Cheong, S. J.**; Park, Y. H. *SPIE*

[IC.5] **A Long-Distance 3D Face Recognition Architecture Utilizing MEMS-Based Region-Scanning LiDAR**
Lim, Y. S.; Lee, S. H.; **Cheong, S. J.**; Park, Y. H. *SPIE*

[DC.1] **Head Pose Estimation Method Using Depth Camera Based on 3D Point Cloud**
**Cheong, S. J.**; Lee, S. H.; Lim, Y. S.; Park, Y. H. *The Korean Society of Mechanical Engineers*

[DC.2] **Validation of Simulation Model for 3D Indirect Time-of-Flight Sensor**
Lee, S. H.; Lim, Y. S.; **Cheong, S. J.**; Park, Y. H. *The Korean Society of Mechanical Engineers*

[DC.3] **Infrared Small Object Detection Based on Histogram Transformation**
**Cheong, S. J.**; Jung, W.; Park, Y. H. *The Korean Society of Mechanical Engineers*

[DC.4] **Depth Image Super Resolution Method for Time-of-Flight Camera Using Single RGB Image Depth Estimation**
Lim, Y. S.; Lee, S. H.; **Cheong, S. J.**; Park, Y. H. *The Korean Society of Mechanical Engineers*

[DC.5] **Depth Error Correction of 3D Scanning Sensor Using Machine Learning**
Lee, S. H.; Lim, Y. S.; **Cheong, S. J.**; Park, Y. H. *The Korean Society of Mechanical Engineers*

## Research & Work Experience

- **University of Wisconsin-Madison** *Aug. 2025 – Current*
  Research Assistant | WISION Lab Madison, WI, USA
  ***Topic: De-glaring in Time-of-Flight Imaging***
  ◦ Contributed to the developement of **de-glaring algorithm** in dToF LiDAR transient domain.

- **LG Innotek Inc.** *Aug. 2023 – Aug. 2025*
  LiDAR Software Researcher | Base Technology Lab Seoul, Republic of Korea
  ***Topic: Single-Photon LiDAR Imaging & Statistical Signal Processing***
  ◦ Contributed to the development of **company's first dToF LiDAR prototype**.
  ◦ Developed **statistical image/signal processing algorithms** for single-photon LiDAR.
  ◦ Designed **artifact-suppression pipelines** for single-photon LiDAR.
  ◦ Developed **range-walk compensation** algorithms for single-photon LiDAR using **statistical signal processing** on time-resolved photon histograms.

- **KAIST** *Sep. 2021 – Aug. 2023*
  Research Assistant | Human-Machine Interaction Lab Daejeon, Republic of Korea
  ***Project: Indirect Time-of-Flight LiDAR Imaging & Signal Processing*** *(Intek Plus Inc., Daejeon, Republic of Korea)*
  ◦ Contributed to the development of an **iToF AMCW LiDAR platform**, including optical architecture, modulation scheme, and signal processing workflow.
  ◦ Developed **machine learning-based signal processing algorithms** to correct **internal stray light and multi-path interference** within the iToF optical system.
  ◦ Built an **iToF LiDAR-based vision platform** that integrates depth and intensity data for downstream 3D perception and human–machine interaction tasks.
  ***Project: Thermal-Infrared Imaging*** *(Korean Coast Guard, Incheon, Republic of Korea)*
  ◦ Led development of an **infrared imaging system** for remote target detection using **physics engine-based simulation** for data augmentation to enhance detectability and localization performance.

## Honors & Awards

- **Summer Research Fellowship** *2026*
  University of Wisconsin-Madison
- **Academic Excellence Scholarship** *2017, 2020*
  Yonsei University
- **Academic Excellence Award** *2017*
  Yonsei University

## Skills

- **Programming Languages:** Python, C/C++, MATLAB, LabVIEW
- **Frameworks and Libraries:** PyTorch, OpenCV, Open3D, MMDetection3D, ROS, Git
- **Visualization Tools:** CloudCompare, MeshLab, Weights and Biases
- **Hardware:** Oscilloscope, Function Generator
- **Language Skills:** Korean (Native), English (fluent)

## Relevant Coursework

- **Machine learning:** Mathematical Foundation of Machine Learning, Advanced Deep Learning in NLP, Pattern Recognition, Theoretical Foundation of Machine Learning, Matrix Methods in Machine Learning, Probability and Statistics, Random Data Analysis and Processing
- **Computer Vision:** Mobile Robotics and Autonomous Navigation, Autonomous Mobile System Programming, Introduction to Visual Intelligence
- **Optimization:** Nonlinear Optimization

## Teaching Experience

- **University of Wisconsin-Madison** *Sep. 2025 – Current*
  Teaching Assistant Madison, WI, USA
  ◦ CS 766: Computer Vision
  ◦ CS566: Introduction to Computer Vision